HiNT-Surv: Hierarchical Niche-Aware Multimodal Survival Prediction from Imaging Mass Cytometry, Clinicopathologic Variables, and Genomic Data

***Abstract*—Accurate breast cancer survival prediction supports prognostic assessment. Cellular organisation and tissue structure provide important prognostic information for breast cancer. Imaging mass cytometry (IMC) offers spatially resolved protein measurements that capture these structural features. However, current methods have not fully used this spatial organizational information for survival prediction. We propose HiNT-Surv, a hierarchical niche-aware framework combining phenotype-aware cell-set encoding, distance-weighted graph message passing, and a niche branch that partitions molecular interaction features using learned neighborhood-composition prototypes. Availability-masked additive fusion integrates IMC with clinical and available genomic evidence under a Cox survival objective. A training-derived cap bounds aggregate IMC dispersion on training patients, while exact branch and core–niche decompositions support interpretation. We evaluated our algorithm on two public datasets. Results across five patient-disjoint outer folds and three seeds gave mean Harrell's C-indices of 0.740 on METABRIC (404 patients) and 0.734 on Basel (285 patients), with the lowest Basel integrated Brier score (IBS) of 0.140 over 24–120 months among the evaluated models. Source-fitted METABRIC-to-Basel transfer without target-cohort retraining yielded a C-index of 0.700 and IBS of 0.150. Exploratory niche analyses supported composition-associated molecular coherence, but evidence for survival-specific refinement remained limited. These findings support competitive multimodal prognosis with inspectable tissue contributions.**

***Index Terms*—Breast cancer, survival prediction, imaging mass cytometry, graph neural networks, multimodal learning, tumor microenvironment.**

I. Introduction

Breast cancer exhibits substantial genomic and transcriptomic heterogeneity, with molecular subgroups associated with distinct clinical trajectories [1], [2]. Accurate survival prediction supports prognostic assessment and risk stratification, providing patient-specific information relevant to treatment planning and long-term follow-up [2], [3]. This motivates models that integrate complementary evidence about the patient, tumor biology, and the surrounding tissue environment.

Established survival approaches include the Cox proportional hazards model [4], while DeepSurv [5] and DeepHit [6] introduce nonlinear risk modeling and discrete event-time prediction, respectively. These methods account for right censoring, but rely on the supplied patient representation to capture prognostically relevant information. Genomic profiles provide molecular evidence beyond routine clinicopathologic variables [1], yet bulk measurements average over heterogeneous tissue and do not preserve spatial relationships among malignant, immune, and stromal cells. Thus, clinical and bulk genomic inputs alone do not explicitly encode local cellular organization.

Imaging mass cytometry (IMC) addresses this measurement gap by quantifying dozens of proteins at single-cell resolution while preserving tissue coordinates [7]. Breast-cancer IMC studies have linked cellular phenotypes, cell-cell interactions, and recurrent microenvironmental structures to genomic features and clinical outcomes [8]–[10]. These findings motivate a multicellular *niche*—a recurring local microenvironment characterized by cellular composition and interaction patterns—as an intermediate representation between individual cells and patient-level prognosis. DMGN integrates region-level IMC features with patient variables for survival prediction [11]. We instead focus on separating nonspatial cellular summaries, graph-derived spatial context, and niche-specific interaction evidence within a multi-core patient representation.

Three challenges arise in translating these spatial measurements into patient-level survival predictions. First, molecular variation, cellular composition, and local spatial context must be represented across multiple tissue cores. Aggregation can conflate these signals, obscuring whether graph propagation and typed cell–neighborhood interactions provide prognostic information beyond nonspatial tissue summaries. Second, cohort-dependent assays, unavailable modalities, and variable tissue coverage complicate multimodal integration, requiring explicit handling of missing inputs and the scale of tissue-derived evidence. Third, spatial representations should support biological interpretation by relating predictions to recurring microenvironment compositions and associated interaction patterns rather than only an opaque aggregate risk score.

To address these challenges, we propose **HiNT-Surv**, a hierarchical niche-aware framework for multimodal survival prediction from IMC, clinicopathologic variables, and genomic data. Our contributions are as follows:

- To capture complementary spatial and molecular information, we develop a modular IMC encoder combining phenotype-aware set encoding, distance-weighted graph message passing, and a niche branch that models phenotype-centered neighborhood interactions through soft assignments to learned composition prototypes. Separate set, graph, and niche risk terms support matched ablations of their incremental predictive value.
- To integrate heterogeneous and incomplete modalities, we formulate additive late fusion of clinical, genomic, and IMC risk terms, with availability masks assigning zero

contribution to missing branches and a training-derived scale cap controlling the IMC correction relative to the clinical–genomic base risk. All active branches are jointly optimized using the Cox partial likelihood with Efron's approximation for tied events.

- To connect predictions with tissue biology, we associate learned niches with niche-specific molecular interaction readouts. The additive formulation provides exact patient-specific contributions across modalities and core–niche pairs, enabling interpretation of the tissue compositions and local interactions driving predicted risk.

We will make the source code for HiNT-Surv publicly available upon acceptance.

II. Related Work

Survival learning. Cox proportional hazards (Cox-PH) models covariate effects on relative risk [4], while DeepSurv replaces the linear predictor with a neural network [5]. DeepHit estimates discrete event-time distributions [6]. Random survival forests [12], BroadSurv [13], and DAGSurv [14] provide alternative approaches based on tree ensembles, broad learning, and structured covariate dependencies; attention-based recurrent models also incorporate longitudinal information [15]. These approaches address censored outcome modeling, but applying them to IMC additionally requires a patient representation that captures cellular organization and combines evidence from multiple tissue cores.

Spatial tissue representation and niches. Spatial models differ in their representation scale and supervision. SPACE-GM learns local cellular subgraphs associated with clinical outcomes [16], while BiGraph links cellular and population graphs to identify prognosis-relevant patient subgroups [17]. NaroNet learns phenotypes, neighborhoods, and larger tissue areas from patient-level labels; its breast-cancer analysis combines multiple images per patient and predicts survival-derived risk groups [18]. CellCharter [19] instead identifies spatial niches through unsupervised modeling of cells and their local environments. These approaches differ in their prediction targets and aggregation strategies, motivating a distinction between discovering tissue patterns and preserving their contributions within a multimodal time-to-event model.

Survival prediction. Multimodal prognosis connects these representations with molecular and clinical evidence. Pathomic Fusion combines histology image and cell-graph representations with genomic features [20]. At single-cell resolution, Yadav et al. combine engineered cellular phenotypes and interaction features with Cox-nnet for breast-cancer survival modeling [21]. DMGN is a directly relevant breast-IMC comparator, using a multimodal graph to combine spatial image-region features with patient-variable embeddings [11]. HiNT-Surv focuses on a complementary design: learned prototypes of local neighborhood composition partition typed molecular interactions, yielding patient-level niche risk contributions explicitly decomposable across niches and cores. Additive late fusion combines clinical, genomic, and IMC evidence, while assay-availability masks and a training-derived scale cap control the aggregate IMC contribution. The patient-level risk is optimized using the Cox partial likelihood with Efron's correction for tied events.

TABLE I
Characteristics of the analytical cohorts.

Characteristic	METABRIC	Basel
Patients, n	404	285
Tissue cores, n	458	376
Deaths, n (%)	186 (46.0)	80 (28.1)
Right-censored, n (%)	218 (54.0)	205 (71.9)
Original IMC panel, n^{a}	37	35

[a]Number of protein biomarkers reported in the original studies, not the number of processed input features. Percentages are calculated using the patient count of each cohort.

III. Method

A. Datasets

We use two publicly available breast cancer cohorts with Imaging Mass Cytometry (IMC), clinical annotations, and survival follow-up: METABRIC [8] and Basel [9]. Table I summarizes the records used in our experiments, not the full populations of the source studies. The endpoint is overall survival (OS), with all-cause death as the event and right censoring at the recorded follow-up time when no death is observed. Patients are the unit of survival analysis, with tissue cores linked to their corresponding patient identifiers.

METABRIC: IMC images and segmentation masks are available through the Image Data Resource (accession `idr0076`) and are used together with the published single-cell measurements. The retained cores contain 406,039 annotated cells. Clinical and survival records are linked through METABRIC patient identifiers. Available gene-expression, copy-number alteration, and somatic-mutation profiles obtained from cBioPortal [22] provide additional molecular inputs for METABRIC, with assay availability recorded explicitly for each patient.

Basel: We use the University Hospital Basel IMC resource released by Jackson et al. [9] and its accompanying patient metadata. The clinical annotations include age, receptor status, treatment history, and survival follow-up; the analyzed sample and outcome counts are reported in Table I.

B. Overview and Patient-Level Inputs

HiNT-Surv (Figure 1 predicts patient-level log-relative risk r_p, with $\lambda_p(t) = \lambda_0(t)\exp(r_p)$. Let $\mathcal{C}_{p\nu}$ denote patient p's observed cores of assay ν, $\mathcal{I}_c$ the cells in core c, and $\mathcal{C}_p^* = \mathcal{C}_{p\nu_*}$ the cores of the designated graph assay. Cell i has protein/morphology features $\mathbf{x}_i$, broad and fine phenotypes b_i, f_i, and spatial coordinates; B is the number of broad phenotypes. Parallel cell-set, graph, and niche branches supply IMC risk terms for additive fusion with clinical and available genomic evidence. Here $[\cdot;\cdot]$ concatenates vectors and $\odot$ denotes elementwise multiplication. Training-only fitting, cohort-specific inputs, and evaluation are specified in Sec. IV.

Clinical and genomic branches. Standardized clinical covariates become 16-dimensional tokens with shared scalar

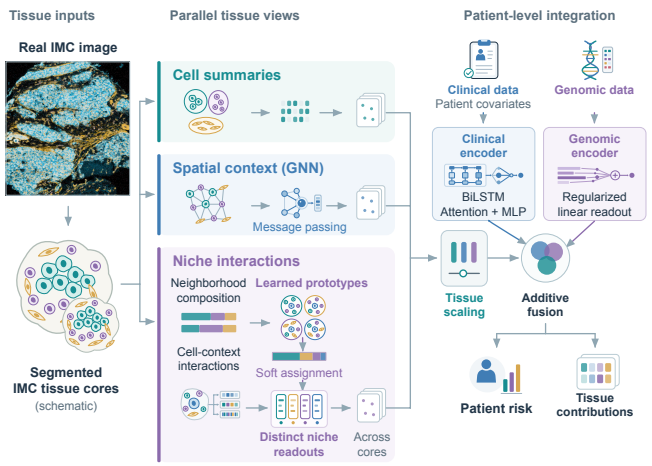


Fig. 1. Overview of HiNT-Surv

projection and learned feature identities. A bidirectional LSTM (24 units per direction), attention pooling, and a $48 \to 32 \to 1$ ReLU head with hidden dropout 0.2 yield r_p^{C}. Tokens represent covariates, not time steps; treatment indicators are fixed recorded covariates. Genomic inputs comprise 59 expression, 59 copy-number alteration (CNA), and 173 mutation features, plus CNA fractions with absolute value at least one/two, $\log(1+\mathrm{TMB})$, and $\log(1+\text{functional panel mutation count})$. G0 uses 12/4/4 expression/CNA/mutation principal components; G1 retains expression, CNA, and recurrent mutations; G2 compresses only G1's expression block to 12 components. All include the burdens. G1/G2 require both mutation values in at least ten training patients. A regularized linear readout gives $r_p^{\mathrm{G}} = \mathbf{w}_G^\top \mathbf{z}_p^{\mathrm{G}}$; the branch is omitted when genomic inputs are excluded.

C. Cell-Set and Spatial Graph Encoders

Cell-set encoding. Residual SiLU stems map $[\mathbf{x}_i; \mathbf{e}^{\mathrm{b}}_{b_i}; \mathbf{e}^{\mathrm{f}}_{f_i}]$ to $\mathbf{h}_i^0 \in \mathbb{R}^{16}$, using four-dimensional phenotype embeddings. Set stems are assay-specific; graph and niche branches share a separate stem. Protein-channel dropout (0.2) shares a mask per patient, assay, and stem, retaining morphology. For $\mathcal{I}_{c,a} = \{i \in \mathcal{I}_c : b_i = a\}$ and $n_{ca} = |\mathcal{I}_{c,a}|$, a type token combines the mean and population dispersion of cell states with a type embedding, cell fraction, $\log(1+n_{ca})/5$, and presence. Absent types are masked; single-cell dispersion is zero. Averaging present-type tokens within cores, then observed cores within patients, yields a scaled, bias-free linear score $r_{p\nu}^{\mathrm{S}}$. An optional bank adds protein quartiles, fractions, log-counts, and presence by phenotype. Quartiles are shrunk toward training-core references with weight $n_{ca}/(n_{ca}+20)$; absent types use the reference. Patient means and inter-core SDs (zero for one core) undergo standardization, 16-component PCA, and rescaling before linear readout $r_{p\nu}^{\mathrm{B}}$.

Spatial graph encoding. Each cell connects to up to eight nearest neighbors within $50\,\mu$m, followed by symmetrization without self-loops; final degrees may exceed eight. For neighbors $\mathcal{N}(i)$, distances d_{ji}, and $\tau \in \{20, 50\}\,\mu$m, define $w_{ji}^\tau = e^{-d_{ji}/\tau}$ and $W_i^\tau = \sum_{j\in\mathcal{N}(i)} w_{ji}^\tau$. Starting from $\mathbf{z}_i^{(0)} = \mathcal{A}_0(\mathbf{h}_i^0)$, two propagation rounds form

$$\begin{aligned}\mathbf{m}_{ji}^{(l)} = M_l([\mathbf{t}_j^{(l)}; \mathbf{t}_i^{(l)}; \mathbf{a}_j^{(l)} \odot \mathbf{a}_i^{(l)};\\ \mathbf{e}^{\mathrm{b}}_{b_j}; \mathbf{e}^{\mathrm{b}}_{b_i}; \boldsymbol{\eta}_{ji}]),\end{aligned} \tag{1}$$

where $\mathbf{t}_i^{(l)}$ is an affine–SiLU transform of $\mathbf{z}_i^{(l)}$, $\mathbf{a}_i^{(l)} \in \mathbb{R}^4$ is a linear projection, and $\boldsymbol{\eta}_{ji} = [d_{ji}/50, (d_{ji}/50)^2, e^{-d_{ji}/50}]$. Weighted message means and population SDs at both bandwidths, the current state, and neighborhood log-counts and presence indicators for each type feed the update network. Isolated-cell moments are zero. Fixed affine normalizers $\mathcal{A}_l$ are fitted sequentially at initialization with equal patient and within-patient cell weights. Concatenated depth-0/1/2 states are projected to eight dimensions, then pooled by phenotype, core, and patient as above to obtain r_p^{Gr}.

D. Niche-Partitioned Molecular Interactions

Molecular interactions. The niche branch uses graph-stem states *before propagation or normalization*. Within core c, states are centered over cells of the same fine phenotype:

$$\widetilde{\mathbf{h}}_i = \mathbf{h}_i^0 - \frac{1}{|\mathcal{I}^{\mathrm{f}}_{c(i),f_i}|} \sum_{j\in\mathcal{I}^{\mathrm{f}}_{c(i),f_i}} \mathbf{h}_j^0, \tag{2}$$

where $\mathcal{I}^{\mathrm{f}}_{c,f} = \{i \in \mathcal{I}_c : f_i = f\}$. For sender type $b \neq b_i$, let $\mathcal{N}_b(i) = \{j \in \mathcal{N}(i) : b_j = b\}$ and $n_{ib} = |\mathcal{N}_b(i)|$. With $\epsilon = 10^{-12}$,

$$\begin{aligned}\mathbf{z}_{ib}^\tau &= \frac{\sum_{j\in\mathcal{N}_b(i)} w_{ji}^\tau \widetilde{\mathbf{h}}_j}{\max\{\sum_{j\in\mathcal{N}_b(i)} w_{ji}^\tau, \epsilon\}},\\ \mathbf{u}_i &= \tanh(U_{b_i}^\top \widetilde{\mathbf{h}}_i), \quad \mathbf{v}_{ib}^\tau = \tanh(V_b^\top \mathbf{z}_{ib}^\tau),\\ \boldsymbol{\ell}_{ib}^\tau &= \frac{n_{ib}}{n_{ib}+4}[\mathbf{u}_i; \mathbf{v}_{ib}^\tau; \mathbf{u}_i \odot \mathbf{v}_{ib}^\tau].\end{aligned} \tag{3}$$

Here $U_b, V_b \in \mathbb{R}^{16\times 2}$; the six-dimensional feature retains receiver/context marginals and their product, attenuates sparse neighborhoods, and vanishes for absent sender types.

Composition-based assignment. Neighborhood fractions across both bandwidths form $\boldsymbol{\pi}_i \in \mathbb{R}^{2B}$, with standardized descriptor $\widetilde{\boldsymbol{\pi}}_i$:

$$\begin{aligned}\pi_{ib}^\tau &= \frac{\sum_{j\in\mathcal{N}(i)} w_{ji}^\tau \mathbb{I}(b_j = b)}{\max\{W_i^\tau, \epsilon\}},\\ q_{im} &= \frac{\exp[-\|\widetilde{\boldsymbol{\pi}}_i - \mathbf{p}_m\|_2^2/(2BT)]}{\sum_{k=1}^M \exp[-\|\widetilde{\boldsymbol{\pi}}_i - \mathbf{p}_k\|_2^2/(2BT)]}.\end{aligned} \tag{4}$$

Eligible receivers have heterotypic neighbors. Standardization gives equal mass to eligible patients, then their eligible cores, then receivers; descriptors are not clipped. Weighted K-means initializes $M = 4$ prototypes using at most 512 eligible receivers per core, with sampling weights preserving patient/core mass. Four distinct sampled descriptors and four occupied clusters are required. Prototypes satisfy $\mathbf{p}_m = \mathbf{p}_m^{(0)} + \boldsymbol{\Delta}_m$, with zero-initialized offsets and $T = 1$. Ineligible receivers

have $q_{im} = 0$. Descriptors and standardization remain fixed while survival learning updates the offsets, without entropy, balance, or spatial-smoothness penalties. Niches need not be spatially contiguous.

Niche-specific readout. For ordered types $a \neq b$,

$$\begin{aligned} \mathbf{\Phi}^{\tau}_{cm,ab} &= \frac{\sum_{i\in\mathcal{I}_{c,a}} q_{im}\boldsymbol{\ell}^{\tau}_{ib}}{\max\{n_{ca},1\}}, \\ r^{\mathrm{N}}_{cm} &= \boldsymbol{\theta}^{\top}_{m}(\mathbf{s}^{\mathrm{N}} \odot \mathbf{\Phi}_{cm}), \qquad r^{\mathrm{N}}_{p} = \frac{\sum_{c\in\mathcal{C}^{*}_{p}}\sum_{m} r^{\mathrm{N}}_{cm}}{|\mathcal{C}^{*}_{p}|}. \end{aligned} \tag{5}$$

The denominator preserves niche mass relative to *all* type-a receivers, rather than normalizing within each niche. Concatenation gives $\mathbf{\Phi}_{cm} \in \mathbb{R}^{12B(B-1)}$. Coefficients factorize as $\theta_{m,\tau abtk} = \eta_{m,t\tau k}A_{m,t\tau ak}B_{m,t\tau bk}$, where t indexes receiver/context/product and $k \in \{1,2\}$; molecular projections are shared across niches. Fixed inverse population-SD readout scales are fitted once with dropout disabled from unassigned, equal-core training-patient features, including zero vectors for absent assays, using an SD floor of 0.01 and clipping to $[0.25, 20]$. Set and graph readouts use the same rule, without centering or bias. Unsupported cores contribute zero but remain in the patient average; missing graph assays give $r^{\mathrm{N}}_{p} = 0$. Since $\sum_m \mathbf{\Phi}_{cm} = \mathbf{\Phi}_c$, the unpartitioned representation, distinct heads make predictions assignment-dependent, whereas shared heads cancel the assignments. $M = 1$ recovers the unpartitioned branch.

E. Additive Fusion and Survival Learning

A linear support head maps standardized assay availability, tissue abundance, phenotype composition, graph-degree summaries, and typed neighborhood support to r^{sup}_{p}. With $o_{p\nu} = \mathbb{I}(|\mathcal{C}_{p\nu}| > 0)$ and bank setting $\beta \in \{0,1\}$,

$$\begin{aligned} r^{\mathrm{base}}_{p} &= r^{\mathrm{C}}_{p} + r^{\mathrm{G}}_{p}, \\ r^{\mathrm{IMC}}_{p} &= r^{\mathrm{sup}}_{p} + \sum_{\nu} o_{p\nu}(r^{\mathrm{S}}_{p\nu} + \beta r^{\mathrm{B}}_{p\nu}) \\ &\quad + o_{p\nu_{*}}(r^{\mathrm{Gr}}_{p} + r^{\mathrm{N}}_{p}), \\ r_{p} &= r^{\mathrm{base}}_{p} + \kappa r^{\mathrm{IMC}}_{p}. \end{aligned} \tag{6}$$

Masks act after assay heads; the support term remains active. For training-patient population SDs σ_{B} and σ_{I} of base and IMC scores,

$$\kappa = \min\left\{1, \frac{0.3\ \mathrm{sg}[\sigma_{\mathrm{B}}]}{\sigma_{\mathrm{I}} + 10^{-8}}\right\}, \tag{7}$$

where sg stops numerator gradients only. Thus $\mathrm{SD}_{\mathrm{tr}}(\kappa r^{\mathrm{IMC}}) \leq 0.3\sigma_{\mathrm{B}}$, a training-dispersion bound, not a held-out guarantee. The current full-batch cap is used during training; after each update it is recomputed on training patients with dropout disabled and stored for inference. For a fitted model, $\kappa r^{\mathrm{N}}_{cm}/|\mathcal{C}^{*}_{p}|$ gives an exact additive core–niche contribution, not a causal effect.

For observed times T_p, event indicators δ_p, and $D = \sum_p \delta_p > 0$, training minimizes

$$\mathcal{L} = -D^{-1}\ell_{\mathrm{Efron}}(\mathbf{r}; \mathbf{T}, \boldsymbol{\delta}) + \|\mathbf{w}_G\|^2_2, \tag{8}$$

where ℓ_{Efron} is the Cox partial log-likelihood with Efron's tied-event correction [4], [23]; the genomic penalty is omitted without genomics. All trainable branches and prototype offsets are jointly optimized using full-batch AdamW and complete patient risk sets, without core-count weighting.

IV. Experimental Protocol

A. Experimental Setup

Evaluation design. Patient-level nested cross-validation was performed separately in METABRIC and Basel using five fixed outer folds, three inner folds per outer-training partition, and seeds $\{0,1,2\}$, shared across methods. All cores and assays of each patient remained in one partition. Each configuration required $45 = 5 \times 3 \times 3$ inner runs per cohort, with candidate checkpoints evaluated along each trajectory; selected models required 15 outer refits per model and cohort. All data-dependent fitting—imputation, scaling, feature selection (including mutation filtering), dimensionality reduction (including PCA), feature-bank references, prototype initialization, activation calibration, and structure estimation—used only the corresponding training partition.

Baseline implementation. Comparators were Cox-PH, RSF, DeepSurv, single-event DeepHit, BroadSurv, DAGSurv, AttentionSurv, and patient-level DMGN under modality-matched evaluation. Tabular baselines received HiNT-Surv's clinical covariates and eligible genomic features plus outcome-independent IMC summaries from the same assays: phenotype-specific protein quartiles, cell-type fractions, log-counts, and presence indicators, aggregated across observed cores by patient means and inter-core standard deviations. Genomics was omitted in Basel; each applicable METABRIC baseline independently selected its genomic representation from the prespecified alternatives.

Cox-PH and RSF used `scikit-survival`; DeepSurv and DeepHit used `pycox`. BroadSurv, DAGSurv, and AttentionSurv were reproduced from their cited formulations. Adapted from the authors' released implementation, DMGN retained its image-region encoder and multimodal graph with matched patient-variable inputs. Its core log-risks were equally averaged before both survival-loss computation and evaluation, yielding one survival record per patient. Implementation sources, versions, architectural adaptations, missing-input handling, and model-specific search spaces and stopping rules are documented in the accompanying implementation specification. All methods selected hyperparameters and applicable training duration by mean inner-validation patient-level Harrell's C and refitted on the full outer-training partition, retaining their own objectives and optimization procedures rather than adopting HiNT-Surv's schedule.

Concordance used native log-risks for Cox-based readouts and ensemble mortality for RSF. Survival probabilities used training-fitted Breslow baseline cumulative hazards for Cox-based models and RSF's native ensemble survival function. DeepHit and DAGSurv used predicted event-time probability masses, with scalar risk $r_p = -\int_0^{120} \widehat{S}_p(t)\,\mathrm{d}t$ fixed for selection and held-out evaluation. Discrete-time cut-points

were training-derived; survival curves were right-continuous step functions. Horizon-specific AUC used $1 - \widehat{S}_p(t)$; IBS used the common monthly grid from 24 to 120 months without renormalizing event probabilities to that interval. All metrics followed the shared censoring-support and fold–seed aggregation rules specified in the evaluation protocol, with equal patient weights ($w_p = 1$).

Training and configuration. For HiNT-Surv, the niche count, assignment temperature, and dispersion-cap coefficient were fixed at $M = 4$, $T = 1$, and $\gamma = 0.3$, respectively. All trainable branches and prototype offsets were jointly optimized by full-batch AdamW with the Cox–Efron objective (Sec. III-E) and complete training-patient risk sets, each patient contributing once irrespective of core count. Learning rates were 2×10^{-3} for clinical and genomic groups and 3×10^{-4} for IMC; nonclinical weight decay was 10^{-3}, and gradient norms were clipped to five per parameter group.

The search combined clinical weight decay $\lambda_C \in \{10^{-4}, 10^{-3}\}$ and feature-bank inclusion $\beta \in \{0, 1\}$ with genomic representations $G \in \{\mathrm{G0}, \mathrm{G1}, \mathrm{G2}\}$ (Sec. III-B) in METABRIC (12 configurations); Basel omitted genomics (four configurations). All other architectural and optimizer settings were fixed. Each configuration was trained for 200 updates per inner fold and seed, evaluating budgets $U \in \{25, 50, 75, 100, 150, 200\}$. A configuration–budget pair was eligible only if all nine Harrell's C scores were defined, each requiring finite risk predictions and at least one comparable patient pair. Selection maximized their equally weighted mean. Censoring estimates and IPCW support checks applied only to IPCW-based secondary metrics. Exact ties favored fewer updates, then lexicographic (G, β, λ_C) order, with $\mathrm{G0} < \mathrm{G1} < \mathrm{G2}$ and ascending numerical values; G was omitted for Basel.

For each outer fold, all training-derived transformations and initialization statistics were refitted on the full outer-training partition. Three models were initialized afresh and trained using the selected configuration and budget without transferring inner-fit weights. Outer-test predictions used these checkpoints with dropout disabled and inference caps computed from the corresponding outer-training patients and frozen for inference. Outer-test outcomes were never used to select configurations, budgets, checkpoints, or seeds.

B. Downstream Analyses

METABRIC ablations reused the full model's outer and inner patient partitions and seeds. Each variant independently selected applicable hyperparameters using the same update grid, primary metric, and tie-breaking rule, followed by three fresh outer refits and outer-test evaluation as above.

Removing IMC eliminated all IMC-derived branches, feature banks, availability and tissue-support descriptors, and the support head, leaving $r_p = r_p^C + r_p^G$. Clinical and genomic ablations removed their respective inputs and branches. Variants retaining IMC recomputed the cap from their training-patient scores with $\gamma = 0.3$ and the remaining base risk (r_p^G without clinical inputs; r_p^C without genomics), then froze it for held-out inference. Removing message passing disabled both propagation rounds but retained graph geometry, a depth-0-only readout, and parallel pre-propagation niche computation. Setting $M = 1$ retained one unpartitioned interaction head; removing the niche branch set $r_p^N = 0$. Removing the niche interaction product excluded $\mathbf{u}_i \odot \mathbf{v}_{ib}^{\tau}$ while retaining receiver and context marginals and learnable niche assignments.

Exploratory biological analyses used full-model outer-refit checkpoints and corresponding held-out patients (15 models per cohort; three eligible seed-specific fits per patient). Cross-fit niche comparisons used training-derived phenotype-composition labels; risk contributions were centered and scaled by each fit's training statistics. Patients, not seed-specific predictions, were the aggregation and resampling units. These analyses were not used for model selection or interpreted as confirmatory evidence of biological or causal effects.

External-transfer protocol. METABRIC-to-Basel comparisons used released Basel records, the recorded variables age, ER status, chemotherapy, and hormone therapy with harmonized coding, and harmonized IMC inputs, excluding genomics and cohort-specific treatment variables from both source fitting and target evaluation. IMC features, phenotype mappings, and encoder architecture were fixed before source selection and applied consistently across cohorts. Temporal equivalence of clinical variables and survival origin was not independently verified. Modality-matched comparisons assess representations and learning procedures rather than architectural effects alone.

Transfer settings and budgets were selected by separate three-fold, event-stratified METABRIC cross-validation using the same seeds, update grid, primary metric, support checks, and tie-breaking rule. Only applicable hyperparameters were searched; candidates were scored by equally averaging nine fold–seed metrics, with preprocessing and initialization statistics fitted within each source-training partition. After selection, these transformations and statistics were refitted on the full METABRIC cohort, and three fresh source models were trained for the selected full-batch update count without transferring cross-validation weights. Parameters, fitted transformations, source-derived inference caps, and Breslow baseline cumulative hazards were frozen for Basel evaluation with dropout disabled, without target fine-tuning or survival recalibration.

Basel outcomes served evaluation only, including reverse Kaplan–Meier censoring estimation for external IPCW metrics, never prediction-model selection, fitting, or survival recalibration. Point estimates averaged three seed-specific metrics. The 95% percentile confidence intervals used 10,000 patient bootstrap samples paired across models and seeds. Each resample recomputed Basel censoring estimates and seed-specific metrics before averaging, keeping source models fixed. Evaluation required source and target follow-up support and finite censoring-survival estimates ≥ 0.05 for every nonzero IPCW contribution. Unsupported metrics were unavailable, without weight clipping, patient exclusion, or interval shortening; undefined bootstrap estimates were recorded and reported. Intervals quantify target-sample uncertainty conditional on the fitted source models.

V. Results

A. Internal Validation and Baseline Comparison

Table II summarizes outer-test performance under the patient-level nested protocol in Sec. IV. On METABRIC, HiNT-Surv led Harrell's C (0.740 versus 0.718 for DMGN), Uno's C@120, and AUC@120, whereas DMGN had marginally higher AUC@60 and lower IBS. On Basel, DMGN led Harrell's C (0.742 versus 0.734 for HiNT-Surv), Uno's C@120, and AUC@60, while AttentionSurv led AUC@120. HiNT-Surv nevertheless achieved the lowest IBS (0.140), compared with 0.156 for BroadSurv, the strongest baseline on this metric. These results indicate cohort- and metric-dependent advantages, with leading discrimination on METABRIC and favorable survival-probability prediction error on Basel, rather than uniform superiority across metrics.

Kaplan–Meier analysis of out-of-fold risk stratification further showed clear separation between predicted low- and high-risk patients in both cohorts (Fig. 2). The separation was significant in METABRIC (log-rank $p = 2.64 \times 10^{-26}$) and Basel ($p = 9.32 \times 10^{-9}$), with the predicted high-risk group exhibiting consistently poorer overall survival. These results complement the fold-wise discrimination metrics by showing distinct survival trajectories across predicted risk groups.

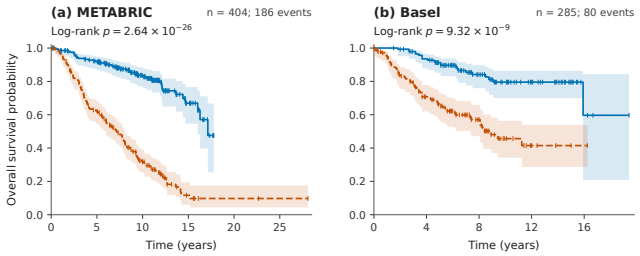


Fig. 2. Kaplan–Meier overall survival curves for HiNT-Surv out-of-fold risk groups in (a) METABRIC and (b) Basel.

B. External Validation

Table III reports source-fitted METABRIC-to-Basel transfer using harmonized clinical and IMC inputs, without genomics or target-cohort retraining or recalibration. HiNT-Surv had the highest point estimates for Harrell's C (0.700), Uno's C@120, and AUC@60, and the lowest IBS (0.150). Its Harrell's C exceeded DeepHit by 0.009, whereas DeepSurv achieved the highest AUC@120 (0.747 versus 0.729 for HiNT-Surv). DMGN showed marked transfer degradation. These results support competitive retrospective performance of the reduced-input model. Bootstrap intervals overlapped substantially for the leading models; between-model differences remain descriptive, not evidence of statistical superiority. The intervals quantify target-sample uncertainty conditional on the fitted source models.

C. Ablation Study

Table IV reports METABRIC ablations, with $\Delta C = C_{\text{variant}} - C_{\text{full}}$. All seven variants had lower mean concordance than the full model. Removing clinicopathologic inputs produced the largest decrease (-0.082), compared with -0.020 without IMC and -0.030 without genomics. Collapsing niches, disabling graph propagation, and removing the niche branch or its interaction product also reduced concordance. Each variant was independently selected and refitted under the same patient partitions and seeds. Removing a base modality additionally changed the reference dispersion used by the IMC cap. The differences therefore reflect complete-system changes rather than additive component importance. They support the combined representation design but do not isolate biological specificity or causal roles.

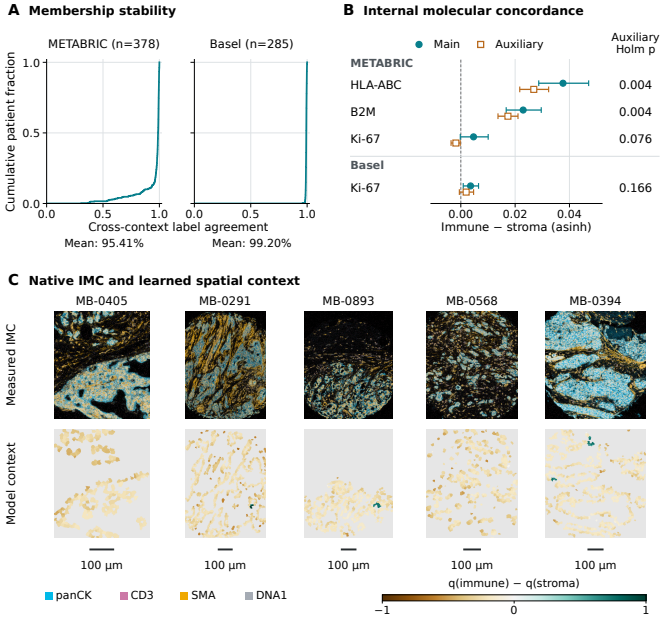


Fig. 3. Biological interpretability of HiNT-Surv. (A) Cross-context niche membership stability in METABRIC and Basel. (B) Internal molecular concordance in main and auxiliary analyses. (C) Native IMC composites and learned spatial context across five METABRIC cores.

D. Biological Interpretability

Figure 3 summarizes niche stability, molecular concordance, and spatial context. Mean cross-context label agreement was 95.41% in METABRIC ($n = 378$) and 99.20% in Basel ($n = 285$; panel A), indicating stable membership across the evaluated contexts.

In METABRIC, HLA-ABC and B2M showed positive immune-minus-stroma contrasts on the asinh intensity scale in both main and auxiliary analyses (panel B; auxiliary Holm-adjusted $p = 0.004$ for each). Ki-67 showed weaker evidence, with auxiliary Holm-adjusted $p = 0.076$ in METABRIC and $p = 0.166$ in Basel. Native IMC composites and corresponding $q_i(\text{immune}) - q_i(\text{stroma})$ maps illustrate the spatial distribution of learned contexts across five METABRIC cores (panel C). Together, these exploratory findings support stable composition-based assignments with internal molecular concordance, but do not establish independent biological validation, causal interactions, or survival-specific mechanisms.

TABLE II
INTERNAL EVALUATION ON METABRIC AND BASEL

Model	Harrell's $C \uparrow$	Uno's C@120 $\uparrow$	AUC@60 $\uparrow$	AUC@120 $\uparrow$	IBS$[24, 120] \downarrow$
A. METABRIC — Internal evaluation					
Cox-PH [4]	0.691 ± 0.020	0.697 ± 0.025	0.704 ± 0.028	0.747 ± 0.062	0.165 ± 0.009
RSF [12]	0.696 ± 0.031	0.700 ± 0.041	0.711 ± 0.031	0.758 ± 0.080	0.163 ± 0.010
DeepSurv [5]	0.709 ± 0.032	0.713 ± 0.042	0.729 ± 0.043	0.761 ± 0.067	0.165 ± 0.009
DeepHit [6]	0.692 ± 0.037	0.694 ± 0.040	0.712 ± 0.035	0.736 ± 0.080	0.313 ± 0.022
BroadSurv [13]	0.704 ± 0.028	0.710 ± 0.032	0.717 ± 0.041	0.767 ± 0.076	0.162 ± 0.012
DAGSurv [14]	0.686 ± 0.025	0.689 ± 0.030	0.708 ± 0.020	0.736 ± 0.073	0.176 ± 0.023
AttentionSurv [15]	0.716 ± 0.024	0.719 ± 0.027	0.733 ± 0.027	0.777 ± 0.070	0.166 ± 0.013
DMGN [11]	0.718 ± 0.027	0.719 ± 0.032	$\mathbf{0.743 \pm 0.027}$	0.762 ± 0.067	$\mathbf{0.160 \pm 0.009}$
HiNT-Surv (Ours)	$\mathbf{0.740 \pm 0.020}$	$\mathbf{0.727 \pm 0.027}$	0.742 ± 0.038	$\mathbf{0.790 \pm 0.065}$	0.165 ± 0.010
B. Basel — Internal evaluation					
Cox-PH [4]	0.698 ± 0.039	0.667 ± 0.060	0.737 ± 0.068	0.696 ± 0.099	0.158 ± 0.012
RSF [12]	0.696 ± 0.036	0.660 ± 0.065	0.743 ± 0.051	0.725 ± 0.103	0.159 ± 0.013
DeepSurv [5]	0.705 ± 0.028	0.667 ± 0.052	0.728 ± 0.050	0.711 ± 0.091	0.167 ± 0.012
DeepHit [6]	0.692 ± 0.042	0.667 ± 0.045	0.721 ± 0.074	0.705 ± 0.083	0.384 ± 0.073
BroadSurv [13]	0.717 ± 0.035	0.678 ± 0.048	0.737 ± 0.058	0.714 ± 0.092	0.156 ± 0.016
DAGSurv [14]	0.699 ± 0.036	0.661 ± 0.058	0.728 ± 0.033	0.700 ± 0.096	0.206 ± 0.038
AttentionSurv [15]	0.723 ± 0.030	0.686 ± 0.047	0.750 ± 0.027	$\mathbf{0.757 \pm 0.098}$	0.166 ± 0.010
DMGN [11]	$\mathbf{0.742 \pm 0.045}$	$\mathbf{0.697 \pm 0.062}$	$\mathbf{0.769 \pm 0.086}$	0.699 ± 0.100	0.167 ± 0.022
HiNT-Surv (Ours)	0.734 ± 0.030	0.685 ± 0.044	0.766 ± 0.067	0.711 ± 0.086	$\mathbf{0.140 \pm 0.013}$

Notes: Values are means $\pm$ sample SDs across five outer folds, after averaging three seed-specific metric values within each fold. Harrell's C uses equal patient weights ($w_p = 1$) for all models.

TABLE III
RETROSPECTIVE EXTERNAL TRANSPORTABILITY FROM METABRIC TO BASEL

Model	Harrell's $C \uparrow$	Uno's C@120 $\uparrow$	AUC@60 $\uparrow$	AUC@120 $\uparrow$	IBS[24,120] $\downarrow$
Cox-PH [4]	0.685 [0.624–0.743]	0.661 [0.598–0.721]	0.732 [0.652–0.806]	0.720 [0.626–0.805]	0.154 [0.134–0.175]
RSF [12]	0.673 [0.605–0.734]	0.654 [0.590–0.717]	0.703 [0.610–0.788]	0.727 [0.635–0.811]	0.153 [0.132–0.175]
DeepSurv [5]	0.689 [0.629–0.744]	0.672 [0.612–0.730]	0.731 [0.653–0.803]	**0.747** [0.657–0.829]	0.152 [0.131–0.173]
DeepHit [6]	0.691 [0.629–0.747]	0.667 [0.605–0.725]	0.668 [0.583–0.750]	0.742 [0.653–0.823]	0.286 [0.271–0.301]
BroadSurv [13]	0.674 [0.613–0.733]	0.653 [0.589–0.714]	0.711 [0.632–0.786]	0.708 [0.613–0.794]	0.160 [0.135–0.185]
DAGSurv [14]	0.674 [0.613–0.733]	0.640 [0.573–0.705]	0.716 [0.635–0.793]	0.685 [0.587–0.777]	0.156 [0.138–0.175]
AttentionSurv [15]	0.654 [0.591–0.715]	0.643 [0.579–0.705]	0.677 [0.595–0.755]	0.693 [0.597–0.782]	0.159 [0.138–0.181]
DMGN [11]	0.485 [0.416–0.552]	0.481 [0.415–0.547]	0.513 [0.428–0.597]	0.559 [0.454–0.661]	0.423 [0.380–0.468]
HiNT-Surv (Ours)	**0.700** [0.624–0.744]	**0.680** [0.608–0.732]	**0.750** [0.647–0.804]	0.729 [0.637–0.814]	**0.150** [0.132–0.183]

Notes: Point estimates average three seed-specific metrics. Brackets show 95% percentile CIs from 10,000 paired patient bootstrap resamples, conditional on fitted source models. Harrell's C uses $w_p = 1$ for all models.

E. Case Study

Figure 4 connects niche assignments, local neighborhoods, and molecular variation in an illustrative IMC core. The native panCK/CD3/SMA/DNA1 composite and HLA-ABC/B2M intensity maps provide spatial reference (A–C), while $q_i(\text{immune}) - q_i(\text{stroma})$ summarizes context for 696 epithelial receivers (D). Four-niche label agreement was 99.56% across 952 eligible cells over six context pairs. Local graphs (E) show phenotype-coded neighborhoods in R1 (189 cells) and R2 (190 cells), with highlighted receiver context scores of -0.211 and -0.166, respectively.

Both markers showed positive immune-minus-stroma contrasts on the asinh intensity scale in all four case comparisons (F). The main ($n = 356$) and auxiliary ($n = 363$) analyses showed the same direction, with larger contrasts for HLA-ABC than B2M (G). Together, the images, receiver-centered graphs, and marker contrasts connect local cellular organization and composition-based assignments with molecular variation. These findings provide descriptive support for HiNT-Surv's spatial interpretability, but do not establish independent biological validation, causal interactions, or survival-specific mechanisms.

VI. DISCUSSION AND CONCLUSION

HiNT-Surv combines multiscale tissue modeling with an additive log-risk formulation, allowing spatial evidence to complement clinical and available genomic inputs while retaining exact branch and core–niche contributions. Nested evaluation yielded the highest mean Harrell's C on METABRIC and the lowest IBS on Basel among the evaluated models. The cohort-dependent rankings indicate complementary strengths in discrimination and survival-probability prediction rather

TABLE IV
HINT-SURV ABLATIONS ON METABRIC

Variant	Harrell's $C \uparrow$ (95% CI)	ΔC (95% CI)
Full HiNT-Surv	**0.740** **[0.701, 0.779]**	—
w/o clinical inputs	0.658 [0.619, 0.697]	−0.082 [−0.104, −0.060]
w/o IMC	0.720 [0.655, 0.785]	−0.020 [−0.038, −0.002]
w/o genomics	0.710 [0.671, 0.749]	−0.030 [−0.042, −0.018]
Niches collapsed ($M = 1$)	0.727 [0.678, 0.776]	−0.013 [−0.023, −0.003]
w/o graph propagation	0.718 [0.679, 0.757]	−0.022 [−0.033, −0.011]
w/o niche branch	0.708 [0.659, 0.757]	−0.032 [−0.036, −0.008]
w/o niche interaction product	0.707 [0.660, 0.754]	−0.033 [−0.037, −0.009]

Notes: $\Delta C = C_{\text{variant}} - C_{\text{full}}$. Variant definitions are given in Sec. IV-B.

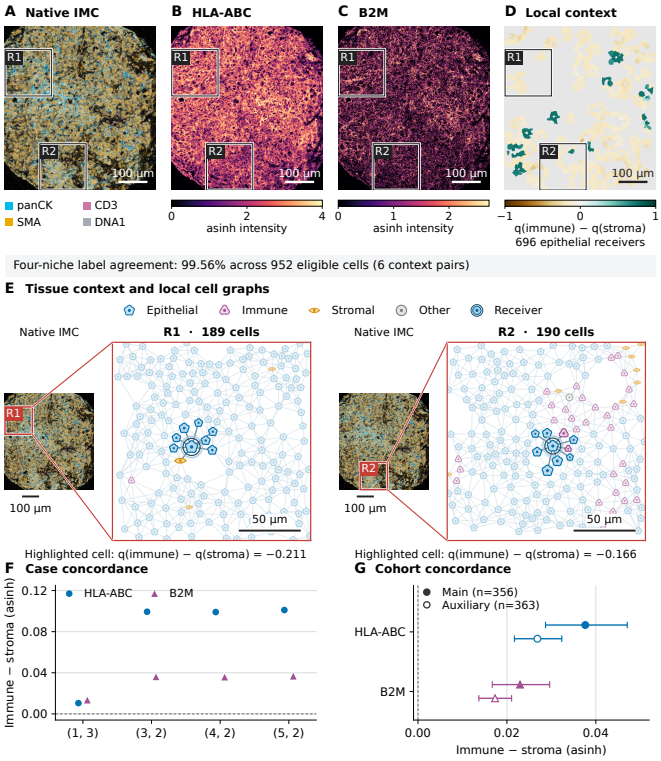


Fig. 4. Local niches and HLA-ABC/B2M molecular variation. (A–C) Native IMC composite and marker intensity maps. (D) Immune-minus-stroma context. (E) Phenotype-coded R1/R2 graphs highlighting receivers and neighbors. (F) Case-level contrasts. (G) Main and auxiliary cohort-level contrasts.

than uniform superiority. Lower IBS alone does not establish calibration or clinical utility.

METABRIC ablations are consistent with genomic and spatial evidence supplementing clinicopathologic information. Removing clinical inputs produced the largest mean concordance decrease, whereas excluding IMC reduced it by 0.020. Lower concordance after collapsing niches, disabling propagation, or removing interaction products supports the combined representation design. However, each variant was independently selected and refitted, and removing a base modality also changed the reference dispersion for the IMC cap. These comparisons therefore measure system-level changes rather than additive component importance or isolated biological mechanisms.

Source-fitted METABRIC-to-Basel transfer achieved a C-index of 0.700 and IBS of 0.150 without target retraining or recalibration. This evaluation used harmonized clinical and IMC inputs without genomics, supporting retrospective transportability of the reduced-input model rather than validating the full METABRIC configuration. Between-model differences remain descriptive, and bootstrap uncertainty is conditional on the fitted source models. A single source–target pairing does not establish general robustness or identify which architectural choices account for transfer performance.

Biological analyses connect stable niche assignments with measurable molecular variation. Positive immune-minus-stroma HLA-ABC and B2M contrasts in METABRIC, together with corresponding case-level patterns, support composition-associated molecular coherence. These internal associations do not independently validate survival-specific biology. Similarly, exact risk decomposition explains how the fitted model combines tissue evidence, not whether the represented cellular interactions are causal. The biological findings should therefore be viewed as exploratory characterization of learned representations rather than mechanistic validation.

Limitations include retrospective cohorts, modest event counts, heterogeneous assays, and incomplete molecular coverage. Temporal equivalence of recorded predictors and survival origins was not independently verified, restricting interpretation to retrospective risk modeling rather than established diagnosis-time prognosis. Availability masks explicitly handle absent assay branches, while the IMC cap bounds training dispersion; neither guarantees robustness to arbitrary missingness or distribution shift. Independent validation should verify predictor-availability windows, assess sensitivity to tissue coverage and missing modalities, and evaluate calibration and clinical utility.

Conclusion. HiNT-Surv integrates phenotype-aware cellular summaries, graph-derived context, and niche-partitioned interactions with clinical and available genomic evidence for patient-level survival prediction. Competitive internal and external results support this representation strategy, while its additive formulation preserves inspectable tissue-risk contributions. The framework provides a structured basis for investigating the prognostic value of multicellular organization, with independent validation required before clinical application.

REFERENCES

[1] C. Curtis *et al.*, "The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups," *Nature*, vol. 486, no. 7403, pp. 346–352, 2012.

[2] O. M. Rueda *et al.*, "Dynamics of breast-cancer relapse reveal late-recurring ER-positive genomic subgroups," *Nature*, vol. 567, no. 7748, pp. 399–404, 2019.

[3] M. J. van de Vijver *et al.*, "A gene-expression signature as a predictor of survival in breast cancer," *The New England Journal of Medicine*, vol. 347, no. 25, pp. 1999–2009, 2002.

[4] D. R. Cox, "Regression models and life-tables," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 34, no. 2, pp. 187–202, 1972.

[5] J. L. Katzman, U. Shaham, A. Cloninger, J. Bates, T. Jiang, and Y. Kluger, "DeepSurv: Personalized treatment recommender system using a Cox proportional hazards deep neural network," *BMC Medical Research Methodology*, vol. 18, no. 1, p. 24, 2018.

[6] C. Lee, W. R. Zame, J. Yoon, and M. van der Schaar, "DeepHit: A deep learning approach to survival analysis with competing risks," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 2314–2321.

[7] C. Giesen *et al.*, "Highly multiplexed imaging of tumor tissues with subcellular resolution by mass cytometry," *Nature Methods*, vol. 11, no. 4, pp. 417–422, 2014.

[8] H. R. Ali *et al.*, "Imaging mass cytometry and multiplatform genomics define the phenogenomic landscape of breast cancer," *Nature Cancer*, vol. 1, no. 2, pp. 163–175, 2020.

[9] H. W. Jackson *et al.*, "The single-cell pathology landscape of breast cancer," *Nature*, vol. 578, no. 7796, pp. 615–620, 2020.

[10] E. Danenberg *et al.*, "Breast tumor microenvironment structures are associated with genomic features and clinical outcome," *Nature Genetics*, vol. 54, no. 5, pp. 660–669, 2022.

[11] X. Fu, E. Patrick, J. Y. H. Yang, D. D. Feng, and J. Kim, "Deep multimodal graph-based network for survival prediction from highly multiplexed images and patient variables," *Computers in Biology and Medicine*, vol. 154, p. 106576, 2023.

[12] H. Ishwaran, U. B. Kogalur, E. H. Blackstone, and M. S. Lauer, "Random survival forests," *The Annals of Applied Statistics*, vol. 2, no. 3, pp. 841–860, 2008.

[13] G. Wu, J. Duan, J. Wang, L. Wang, C. Dong, and C. Lv, "BroadSurv: A novel broad learning system-based approach for survival analysis," in *Proceedings of the 2021 8th International Conference on Information, Cybernetics, and Computational Social Systems (ICCSS)*. IEEE, 2021, pp. 26–31.

[14] A. K. Sharma, R. Kukreja, R. Prasad, and S. Rao, "DAGSurv: Directed acyclic graph based survival analysis using deep neural networks," in *Proceedings of the 13th Asian Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 157. PMLR, 2021, pp. 1065–1080. [Online]. Available: https://proceedings.mlr.press/v157/sharma21a.html

[15] Z. Sun, W. Dong, J. Shi, K. He, and Z. Huang, "Attention-based deep recurrent model for survival prediction," *ACM Transactions on Computing for Healthcare*, vol. 2, no. 4, pp. 35:1–35:18, 2021.

[16] Z. Wu *et al.*, "Graph deep learning for the characterization of tumour microenvironments from spatial protein profiles in tissue specimens," *Nature Biomedical Engineering*, vol. 6, no. 12, pp. 1435–1448, 2022.

[17] Z. Wang, C. A. Santa-Maria, A. S. Popel, and J. Sulam, "Bi-level graph learning unveils prognosis-relevant tumor microenvironment patterns in breast multiplexed digital pathology," *Patterns*, vol. 6, no. 3, p. 101178, 2025.

[18] D. Jiménez-Sánchez, M. Ariz, H. Chang, X. Matias-Guiu, C. E. de Andrea, and C. Ortiz-de-Solórzano, "NaroNet: Discovery of tumor microenvironment elements from highly multiplexed images," *Medical Image Analysis*, vol. 78, p. 102384, 2022.

[19] M. Varrone, D. Tavernari, A. Santamaria-Martínez, L. A. Walsh, and G. Ciriello, "CellCharter reveals spatial cell niches associated with tissue remodeling and cell plasticity," *Nature Genetics*, vol. 56, no. 1, pp. 74–84, 2024.

[20] R. J. Chen *et al.*, "Pathomic Fusion: An integrated framework for fusing histopathology and genomic features for cancer diagnosis and prognosis," *IEEE Transactions on Medical Imaging*, vol. 41, no. 4, pp. 757–770, 2022.

[21] S. Yadav, S. Zhou, B. He, Y. Du, and L. X. Garmire, "Deep learning and transfer learning identify breast cancer survival subtypes from single-cell imaging data," *Communications Medicine*, vol. 3, no. 1, p. 187, 2023.

[22] E. Cerami *et al.*, "The cBio cancer genomics portal: An open platform for exploring multidimensional cancer genomics data," *Cancer Discovery*, vol. 2, no. 5, pp. 401–404, 2012.

[23] B. Efron, "The efficiency of Cox's likelihood function for censored data," *Journal of the American Statistical Association*, vol. 72, no. 359, pp. 557–565, 1977.